Using deep learning to generate synthetic tax records

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What do we want?

Augmented tax unit microdata
for every year, including future years

What do we have?

- **One year of tax data, filers only**
  - About ten years out of date
  - Some values are blurred and some records are distorted
  - No non-filers, and none of the other variables a tax calculation needs: consumption, wealth, unrealized gains, estates
- **Surveys** that hold some of this information, but noisily and not at administrative quality
- **Aggregate targets**
  - Past years: IRS totals by AGI group, filing status and age range
  - Future years: CBO-style projections of 1040 items, or of total wages, business profits and similar totals

The task

Turn these scattered sources into

synthetic, realistic, augmented tax unit microdata

for every year

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m

1. **Start with the IRS PUF**, whatever year is available.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Billionaires	Married	1	$1.2m	$14m	$310m
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m
Non-filers	Single	2,310	$0	$0	$0
	Single	1,870	$6.5k	$0	$0

2. **Add records**: non-filers, and the very rich the PUF leaves out.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	812	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,404	$38k	$0	$0	$9k	$34k	$6.1k
	Married	655	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,120	$51k	$12	$0	$31k	$47k	$0
	Single	3	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,310	$0	$0	$0	$2k	$19k	$0
	Single	1,870	$6.5k	$0	$0	$1k	$15k	$1.2k

3. Add columns: estimate Y given X in a survey, where X is also on the PUF; draw Y for each record.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	860 ↑	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,290 ↓	$38k	$0	$0	$9k	$34k	$6.1k
	Married	702 ↑	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,180 ↑	$51k	$12	$0	$31k	$47k	$0
	Single	2 ↓	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,450 ↑	$0	$0	$0	$2k	$19k	$0
	Single	1,760 ↓	$6.5k	$0	$0	$1k	$15k	$1.2k

4. **Reweight to the target year**: grow weights for population and dollars for prices, then move each weight up or down until the year's totals are hit.

Made-up records.

What were the problems with this approach?

- **It did not work well for past years.** The affordability project made this clear.
- **A separate model for each variable**, each estimating its conditional relationships in its own way.
- **Reweighting worked for only a few past years.** Tax law changed what the IRS targets measure, mainly TCJA cutting back itemized deductions.
- **Stretched further every year.** Plausible records for future years, but worse with each year away from 2015.

The ideal approach

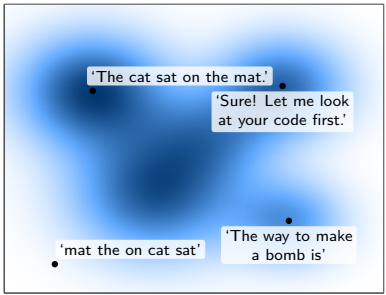
1. Take in survey data covering the joint relationships among every variable we might want to model.
2. Learn all of those relationships in **One Huge Joint Probability Distribution**.
3. For any target year, supply the targets that describe that year, and pull out of it the records that match them.

The result: a realistic synthetic microdata set for that year.

From a conversation with John and Harris.

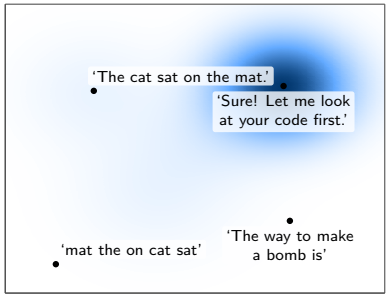
Language models: pretraining, then post-training

Pretraining



Read the whole internet and learn which sentences are likely: a prior over language

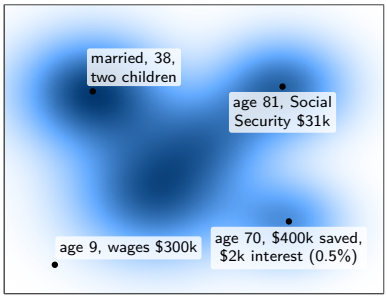
Post-training



Squeeze the mass toward a helpful, harmless assistant, and away from nonsense and harm

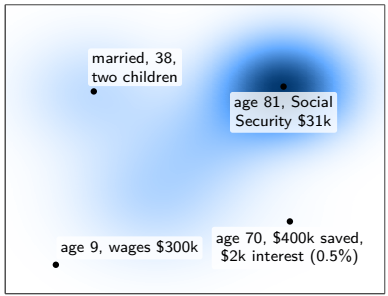
Our model: pretraining, then post-training

Pretraining



Learn which records are likely, from every source: a prior over records

Post-training for 2040



Squeeze the mass toward records that hit 2040's totals, each still coherent

Post-training finds the distribution closest to the prior that hits the year's totals.

From one model per variable to one model for every question

Before (slide 4)

A separate model of Y given X for each imputed variable, each with its own X and form:

- net worth given PUF items, fitted on the SCF
- consumption given PUF items, fitted on the CEX
- tips given PUF items, fitted on another survey

Now

Any joint distribution is a chain of conditionals:

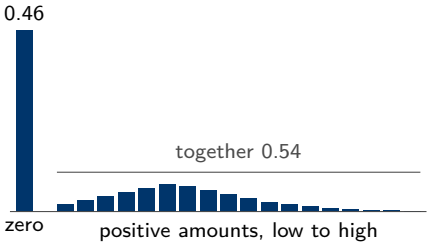
$$
\begin{aligned}
&f(\text{married}, \text{wages}, \text{interest}, \text{home value}) \\
&\quad = f(\text{married}) \\
&\quad \times f(\text{wages} \mid \text{married}) \\
&\quad \times f(\text{interest} \mid \text{married}, \text{wages}) \\
&\quad \times f(\text{home value} \mid \text{married}, \text{wages}, \text{interest})
\end{aligned}
$$

This holds in any order of the variables.

One model that answers every "this variable, given those" question therefore holds the whole joint distribution.

What the model returns: a probability for every range

Question: interest, given source PUF, year 2015, married, wages in the $142k range



Made-up probabilities.

- Each dollar amount is one of **84 ranges**: zero, 10 of losses, 73 of positive amounts. Ranges are percentile groups, finer at the top (each top range holds 0.01% of holders).
- The answer is a **full distribution** over the ranges, like a multinomial logit, not a point prediction.
- In: the source (PUF, SCF, CEX, CPS or SIPP), the year, and whatever is known about the unit.
- One flexible function with about 1.4 million coefficients answers every question.

Fitting it: maximum likelihood on randomly chosen questions

1. Take a real record: PUF, SCF, CEX, CPS or SIPP. Its source and year are always shown.
2. Hide a random share of its variables, anywhere from one to all of them.
3. For each hidden variable, the model returns a distribution given the variables still shown.
4. Score: the log of the probability it put on the true range.
5. Move the coefficients to raise the weighted sum of scores over all records.

New random questions on each of 80 passes through the data, so the one model learns every "this given those".

$$\max_{\theta} \sum_{i} w_i \sum_{j\ \text{hidden}} \log p_\theta\big(x_{ij} \mid \text{shown}_i\big)$$

- θ: the coefficients; x_{ij}: record i's true range for variable j.
- w_i: survey weight, scaled so the few top-income records count as much as the many in the middle.
- A variable a source never measures (the balance sheet on the PUF) is never shown or asked.
- 2% of records held back to check the fit.

Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		

Each step: the model's distribution given everything drawn so far; draw one range from it.

Made-up numbers. The PUF items come in a random order for each record.

Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		
married, given the above		**yes**

Each step: the model's distribution given everything drawn so far; draw one range from it.

Made-up numbers. The PUF items come in a random order for each record.

Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		
married, given the above		**yes**
wages, given all of the above		**$142k range**

Each step: the model's distribution given everything drawn so far; draw one range from it.

Made-up numbers. The PUF items come in a random order for each record.

Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		
married, given the above		**yes**
wages, given all of the above		**$142k range**
interest, given all of the above		**$310 range**

Each step: the model's distribution given everything drawn so far; draw one range from it.

Made-up numbers. The PUF items come in a random order for each record.

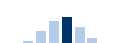
Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		
married, given the above		**yes**
wages, given all of the above		**$142k range**
interest, given all of the above		**$310 range**
home value, as the SCF would see this unit		**$450k range**

Each step: the model's distribution given everything drawn so far; draw one range from it.

Made-up numbers. The PUF items come in a random order for each record.

Writing a record: draw down the chain

Ask the model	Its answer	Draw
Set first: source PUF, year 2015, AGI group p95–99		
married, given the above		**yes**
wages, given all of the above		**$142k range**
interest, given all of the above		**$310 range**
home value, as the SCF would see this unit		**$450k range**
food at home, as the CEX would see it		**$7.9k range**

Then each range becomes dollars: a real record's amount in that range, grown to the target year. Totals such as AGI are added up from the drawn parts.

Made-up numbers. The PUF items come in a random order for each record.

The stack of drawn records

- **Filers**: 400,000 records, drawn as the PUF would see them, a set number in each of 12 AGI groups, in proportion to the PUF's records in each.
  - Weight of a record = its group's returns ÷ records drawn in the group. This undoes the extra weight the top got in fitting.
- **Non-filers**: 54,379 records, drawn as the CPS would see them.
- **Survey items on every record**: balance sheet as the SCF would see the unit, spending as the CEX would, ages and person items as the CPS would, tips as the SIPP would.
- Every record is a fresh draw from the joint distribution, not a copy of any one survey household.

The latest run, a 2019 file.

Post-training: the smallest change to the weights that hits the targets

Keep every record as drawn. Choose new weights w_i close to the starting weights w_i^0:

$$\min_w \sum_i w_i \log \frac{w_i}{w_i^0}$$ such that every target is within its published value $\pm$ 1 standard error

- The same distance as raking, and the answer has raking's form, one multiplier per target: $w_i = w_i^0 \exp(\lambda_1 x_{i1} + \lambda_2 x_{i2} + \dots)$, where x_{it} is record i's contribution to target t.
- Half a standard error for the SCF wealth targets.
- 2019: **753 targets**: IRS tables by AGI, filing status and age; business income; W-2 tips; non-filer counts; SCF wealth; CEX spending.

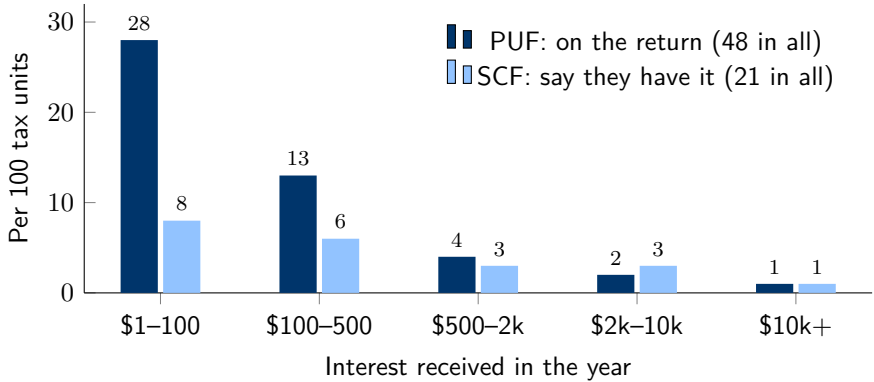
753 of 753
targets inside their ranges

0.05 to 3.2
new weight over old, 1st to 99th percentile

87%
effective sample kept

Latest 2019 run. In the distance, weights are shares of the total.

Challenge: surveys and returns measure differently



The gap is all in small amounts: they go unreported in the survey. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

The fix: two slots and a fill-in

SCF household	interest, as said	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

The answer given is kept. A second slot holds what the return would show, the only slot the PUF has.

Per 100 couples	PUF	SCF	Fill in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

The PUF sets how many are filled in, and at what sizes. The rest of the SCF record sets which households.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted. Rows may not add because of rounding.

Other challenges

- **Building tax units from surveys.** Surveys count households. Each one is split into tax units by the same rules for every survey, and then we decide who files.
- **The return itself changes.** TCJA cut back itemizing; new items, like the tips and overtime deductions, are on no past return.
- **A thin top.** Few very rich records in any source.
- **Future years.** No microdata, only a handful of projected totals.
- **Checking the result.** No source has every variable, so some relationships have nothing to check against.

Where we are

- A 2019 file: 400,000 filer records, plus non-filers
- Five sources: the 2015 PUF, the SCF, the CEX, the CPS and the SIPP, every survey split into tax units by the same rules
- Every PUF item, plus a balance sheet, consumption, tips and auto loan interest
- About 750 targets, all met within their standard errors: IRS tables by AGI, filing status and age, the top 1%, 0.1% and 0.01%, W-2 tips, SCF wealth

What is left

- Rules applied after drawing: credit eligibility, death probabilities, unrealized gains, the basis of capital gains
- The file Tax-Simulator reads, and a first run of Tax-Simulator on it
- A tax calculator inside post-training, for targets on computed tax
- Other years: 2015 to 2023, then the projection years
- The very top: the PUF's aggregate records and the Forbes 400

Questions